

CASE R2 – Institutional Ownership Concentration and U.S. Single-Family Home Price Growth

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Data: Federal Reserve Economic Data (FRED) — CSUSHPISA, FEDFUNDS, CPIAUCSL, UNRATE, GDPC1, HOUST

Question & Mechanism

Do institutional investors push up single-family home prices? We expect institutional buying to exert upward pressure on home prices through a demand concentration channel: large-scale buyers acquire inventory systematically, reduce effective supply for owner-occupants, and are less sensitive to financing costs than individual buyers — meaning their demand persists even as mortgage rates rise. Monetary policy should work in the opposite direction; higher rates raise financing costs, compress affordability, and dampen demand from rate-sensitive buyers. The net price effect in any given year reflects the balance between these two forces. We therefore test whether institutional ownership share is positively associated with home price growth after controlling for the federal funds rate, inflation, unemployment, GDP growth, and new housing supply over the period 2001 to 2024.

Data & Series

I assemble 24 annual observations (2001–2024) from FRED. The dependent variable is the calendar-year average of monthly year-over-year changes in the S&P/Case-Shiller U.S. National Home Price Index (**CSUSHPISA**). The institutional ownership variable (**InstOwn**) is an estimated series constructed from published academic and industry sources, ranging from approximately 1.5% pre-GFC to an estimated 8.5% by 2024, consistent with Pew Research Center (2024) and CoreLogic published estimates. Because no free public FRED series tracks this directly, the estimation is a documented limitation. Macroeconomic controls are sourced directly from FRED:

FEDFUNDS	Federal Funds Effective Rate (monthly avg, annualized)
CPIAUCSL	CPI All Urban Consumers – YoY % change (monthly, SA)
UNRATE	Unemployment Rate (monthly avg)
GDPC1	Real GDP – YoY % change (quarterly, chained 2017\$)
HOUST	New Housing Starts (monthly avg, converted to millions of units)

All series are resampled to annual frequency using calendar-year averages. The regression model is estimated by OLS with HC3 heteroskedasticity-robust standard errors, following MacKinnon and White (1985), to account for the changing variance of residuals across the four macroeconomic regimes in the sample: the pre-GFC expansion, the GFC and recovery, the low-rate iBuyer era, and the pandemic price surge. The full model specification is:

$$HPG_i = \beta_0 + \beta_1 InstOwn_i + \beta_2 FedFunds_i + \beta_3 Inflation_i + \beta_4 Unemployment_i + \beta_5 GDP_Growth_i + \beta_6 HousingSupply_i + \varepsilon_i$$

A restricted model dropping inflation and GDP growth is also estimated as a robustness check.

Table 1: Descriptive Statistics — Annual FRED Data, 2001–2024

Variable	Mean	Std. Dev.	Min	Median	Max	N
HPG (%)	5.02	6.77	-9.35	5.79	16.99	24
InstOwn (%)	3.64	2.14	1.56	2.94	8.50	24
FedFunds (%)	1.74	1.82	0.08	1.24	5.14	24
Inflation (%)	2.55	1.65	-0.31	2.37	8.00	24
Unemployment (%)	5.77	1.84	3.63	5.31	9.61	24
GDP Growth (%)	2.15	1.77	-2.56	2.53	6.27	24
Housing Supply	1.30	0.43	0.55	1.32	2.07	24

Result

Markets responded broadly in the hypothesized directions. On the primary variable of interest, institutional ownership is positively associated with home price growth: a one-percentage-point increase in the institutional share is associated with a **+0.769 pp** increase in annual home price growth ($p = 0.095$), significant at the 10% level with a 95% confidence interval of $[-0.134, +1.672]$. The confidence interval crosses zero, reflecting genuine uncertainty driven by the small sample and the estimated nature of the institutional ownership series.

The federal funds rate is the strongest and most precisely estimated variable in the model: a one-percentage-point increase in the fed funds rate is associated with a **-1.646 pp** change in home price growth ($p = 0.004$, CI: $[-2.771, -0.521]$). This is consistent with the interest rate transmission channel, where higher rates compress affordability and dampen demand from mortgage-rate-sensitive buyers. In the restricted model, the fed funds coefficient strengthens further to -1.907 ($p < 0.001$), confirming it as the dominant driver.

New housing supply enters with a positive and highly significant coefficient of **+13.524** ($p < 0.001$), the opposite sign from the theoretical prior. This is most likely attributable to simultaneity bias: developers respond to rising prices by initiating new construction, meaning starts are elevated precisely in years when prices are already appreciating. OLS cannot disentangle this reverse causality without instrumental variables or lagged regressors. Inflation, unemployment, and GDP growth are all insignificant in the full model, likely due to limited degrees of freedom (17 residual df with 24 observations and 6 regressors) and moderate inter-correlations among controls.

Table 2: OLS Regression Results — Full Model (HC3 Robust SEs), 2001–2024

Variable	Coefficient	Std. Error	z-stat	p-value	Sig.
Constant	-16.339	7.399	-2.208	0.027	**
Inst. Ownership	+0.769	0.460	1.670	0.095	*
Fed Funds Rate	-1.646	0.574	-2.867	0.004	***

Variable	Coefficient	Std. Error	z-stat	p-value	Sig.
Inflation (CPI)	+0.230	1.109	0.207	0.836	
Unemployment	+0.209	0.643	0.326	0.745	
GDP Growth	+0.986	0.610	1.617	0.106	
Housing Supply	+13.524	2.239	6.040	0.000	***

$R^2 = 0.849$ Adj. $R^2 = 0.795$ $F = 17.44$ ($p < 0.001$) $n = 24$ Durbin-Watson = 1.518 *** $p < 0.01$ ** $p < 0.05$ * $p < 0.10$

Table 3: Restricted Model — Robustness Check

Variable	Coefficient	Std. Error	z-stat	p-value	Sig.
Constant	-12.787	9.369	-1.365	0.172	
Inst. Ownership	+0.871	0.503	1.730	0.084	*
Fed Funds Rate	-1.907	0.366	-5.211	0.000	***
Unemployment	-0.230	0.826	-0.279	0.780	
Housing Supply	+14.895	2.848	5.230	0.000	***

$R^2 = 0.795$ Adj. $R^2 = 0.752$ $F = 17.11$ ($p < 0.001$) $n = 24$ *** $p < 0.01$ * $p < 0.10$

Interpretation

The observed sign pattern broadly matches expectations. Institutional ownership is positively associated with home price growth, and the federal funds rate is negatively associated — together accounting for the two primary demand-side forces in the model. The institutional ownership coefficient of approximately 0.77 to 0.87 across specifications implies that the estimated increase in institutional share from roughly 2% in 2012 to 8.5% by 2024 is associated with approximately 5 percentage points of cumulative additional price appreciation over that period, after controlling for monetary policy and macro conditions. This is economically meaningful even if the statistical confidence interval is wide.

The monetary policy finding is the most robust result in the paper. The fed funds rate coefficient of -1.646 (full model) and -1.907 (restricted model) aligns precisely with interest rate transmission theory and confirms that the post-2022 hiking cycle was the dominant force behind the moderation in price growth observed in 2023 and 2024. The 2001 to 2024 window captures four regime shifts — dot-com recovery, GFC, ZIRP era, and pandemic surge — which is why the rate variable carries outsized explanatory power relative to the other controls.

The housing supply sign reversal and the marginal significance on institutional ownership both point to the same underlying constraint: with only 24 annual observations and an estimated key variable, the model cannot separate structural supply-demand relationships from the macroeconomic cycles that drive them simultaneously. The correct interpretation is that institutional ownership is *associated* with higher prices, not that it *causes* them. A causal claim would require MSA-level panel data with directly observed

institutional ownership, lagged regressors or instrumental variables to address simultaneity, and a broader set of structural controls including local zoning restrictions and credit conditions. The present study establishes a directionally plausible first-pass empirical argument and a replicable analytical framework for that future work.

Note on Data

All macroeconomic controls are sourced directly from the Federal Reserve Economic Data (FRED) database and represent actual observed series. The institutional ownership variable is the only estimated series in the model, constructed from Pew Research Center (2024) aggregate estimates and corroborating academic sources. In a live implementation, this series would be replaced with directly observed data from CoreLogic, ATTOM, or Zillow's investor purchase share dataset, which covers the post-2012 period at the metro level. Replacing the estimated series with observed data is the single highest-impact methodological upgrade available to this study.

References

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